

Multiple Choice (10 marks)

1. Calculate the absolute pressure for a system, given a gauge pressure of 1.5 MPa and a barometric pressure (atmospheric pressure) of 104 kPa.
 - (a) 1504 kPa
 - (b) 1.5 MPa
 - (c) 1.604 MPa
 - (d) 1.4 MPa
 - (e) 604 kPa
2. For a dc machine shunt resistance and armature resistance values are
 - (a) high and high
 - (b) high and low
 - (c) low and low
 - (d) low and high
3. The solid angle subtended by an area of 2400 cm^2 on the surface of a sphere of diameter 1.2 m is
 - (a) $3 / 4$.
 - (b) $2 / 5$.
 - (c) $1 / 2$.
 - (d) $1 / 3$.
 - (e) $1 / 5$.
4. Given the probability distribution of the random variable X in the table below, compute $E(X)$ and $\text{Var}(X)$.

x_i	$\Pr(X = x_i)$
0	$8/27$
1	$12/27$
2	$6/27$
3	$1/27$

 - (a) $E(X) = 3/2, \text{Var}(X) = 1/4$
 - (b) $E(X) = 2, \text{Var}(X) = 1/3$
 - (c) $E(X) = 5/3, \text{Var}(X) = 5/9$
 - (d) $E(X) = 1, \text{Var}(X) = 2/3$
 - (e) $E(X) = 3/4, \text{Var}(X) = 3/8$
5. Which of the following windings are necessary in case of all dc machines?
 - (a) star winding
 - (b) delta winding

- (c) closed winding
- (d) parallel winding
- (e) zigzag winding

True / False (10 marks)

Answer True or False

6. True or False: The Laplace inverse of $F(s) = [(5s^2 - 15s + 7) / \{(s + 1)(s - 2)^3\}]$ is $-e^{-t} - e^2 t + 2te^2 t - (t^2 / 2)e^2 t$.
7. True or False: A steel bar designed to withstand an axial load of 5500 lbs with a factor of safety of 2.5 and length of 222 in requires a diameter of 1 5/8 inches.
8. True or False: The variance of a random variable X that is uniformly distributed over the interval $[0, 3]$ is $3/4$.
9. True or False: The carrier swing and frequency deviation of a TV signal with 80% audio modulation are 20 kHz and 40 kHz, respectively.
10. True or False: All types of oscillators, including the Hartley oscillator, generate sinusoidal waveforms in their operation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of finding the Laplace transform of the function $f(t) = t^n$, where n is a positive integer. Explain the significance of the result in engineering applications.
12. Using the provided data for 29 gage sheet steel designated as U.S.S. Transformer 52, calculate the hysteresis and eddy-current losses at the specified frequencies and analyze their implications in engineering applications.

End of Paper